Software Testing Techniques & Models - 1-Page Cheat Sheet

1️⃣ **Equivalence Partitioning (EP)** - **Idea:** Test one representative value from each class. - **Use:** Reduces number of test cases. - **Example:** Tax logic with multiple inputs.

2️⃣ **Boundary Value Analysis (BVA)** - **Idea:** Test edges of input ranges. - **Use:** Detects boundary errors. - **Example:** Input 1–100 → test 0,1,100,101.

3️⃣ **State Transition Testing (STT)** - **Idea:** Test state changes and transitions. - **Use:** Systems with workflows or state machines. - **Example:** ATM states, Traffic lights.

4️⃣ **Decision Table Testing** - **Idea:** Table of rules and conditions; test combinations. - **Use:** Complex conditional logic. - **Example:** Forms with multiple fields, loan approval system.

5️⃣ **Pairwise Testing** - **Idea:** Test all pairs of input combinations. - **Use:** Covers interactions with fewer tests. - **Example:** Checkout flow with payment and discounts.

6️⃣ **Random / Ad-hoc / Monkey Testing** - **Random:** Inputs chosen randomly. - **Ad-hoc:** Informal, unplanned testing. - **Monkey:** Automated random GUI/stress testing.

7️⃣ **Pytest Exception Testing** - **Syntax:**

```
with pytest.raises(ValueError):
    func(-1)
```

- **Use:** Validate that exceptions are raised correctly.

8️⃣ **Pytest Fixtures with yield** - **Idea:** Setup → yield → teardown. - **Use:** Tests with multiple configurations. - **Structure:**

```
@pytest.fixture
def setup_teardown():
    # setup
    yield
    # teardown
```

9️⃣ **Other Models** - **Control Flow Graph:** Program execution paths (white-box). - **Domain Model:** System entities & relationships. - **Exhaustive Testing:** Test all inputs (rarely practical). - **Smoke Testing:** Quick high-level run to check system works.

Quick Memory Table | Technique / Model | Key Idea | Example | |----------|---------|---------| | EP | One value per class | Tax logic | | BVA | Test edges | Input 1–100, test 0,1,100,101 | | STT | Test state changes | ATM, Traffic lights | | Decision Table | Combinations of rules | Forms, loan approval | | Pairwise | All pairs of inputs | Checkout options | | Random / Ad-hoc / Monkey | Random inputs | Stress testing | | Pytest Exception | Check exceptions | ValueError test | | Fixture with yield | Setup/teardown per config | Parametrized tests |